

D873

Laboratory Safety

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher ordered a new solvent in a 4-liter amber jug through the district's new online ordering portal. Before the first lab, she needs to know which glove material and eye protection to require. Where on the SDS should she look?

- A. Section 4, First-Aid Measures
- B. The bottle label, which always lists all required PPE
- C. Section 2, Hazards Identification
- D. Section 8, Exposure Controls/Personal Protection

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A container of potassium nitrate, also sold at garden centers as a fertilizer, carries a pictogram of a flame above a circle. What hazard does the pictogram indicate?

- A. It is an oxidizer that can intensify a fire, so store it away from flammables.
- B. It is a flammable solid.
- C. It is an explosive.
- D. It is low-risk, because it is sold as a garden fertilizer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A chemical is classified as a skin and eye irritant with low acute toxicity. Which pictogram should appear on its label?

- A. Corrosion
- B. Health hazard (person with starburst)
- C. Exclamation mark
- D. Skull and crossbones

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

During a titration lab, a learner gets a splash of dilute acid in one eye. The eyewash station was inspected last Monday and is 6 meters away. After guiding the learner to the eyewash, what should the teacher do?

- A. Pat the eye dry with a paper towel and send the learner back to work.
- B. Flush the eye for at least 15 minutes, holding the lids open, then report and arrange medical follow-up.
- C. Flush for a minute or two until the stinging stops.
- D. Rinse the eye with a mild base to neutralize the acid.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher with 15 years of experience and a regional teaching award did not require goggles during a lab, and a learner's eye was injured by a splash. A family claims negligence. Which elements must be shown for the claim to succeed?

- A. Nothing more; any injury in a lab is automatically negligence.
- B. Only that the teacher breached a safety rule
- C. That the teacher intended to cause harm
- D. A duty of care, a breach of it, the breach causing the injury, and an actual injury

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A teacher failed to post lab safety rules, but no one was ever hurt. Can negligence be established?

- A. Yes, because teachers always owe a duty of care.
- B. No, because teachers are always immune from lawsuits.
- C. No, because there was no injury.
- D. Yes, because a breach of duty occurred.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A combined lab-classroom of 1,200 square feet has two fume hoods installed in 2018 and windows along one wall. Thirty learners are enrolled in the section. Using the guideline of 60 square feet per learner, how many learners should work in the room during labs?

- A. 20
- B. 26
- C. 30
- D. 24

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A school wants a combined lab-classroom that safely holds 24 learners at 60 square feet each. What is the minimum floor area?

- A. 84 square feet
- B. 1,080 square feet
- C. 60 square feet
- D. 1,440 square feet

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A science department still uses mercury thermometers bought in bulk years ago, and one broke last week near a heating vent. What control best reduces the hazard going forward?

- A. Train learners on mercury spill cleanup.
- B. Require nitrile gloves whenever thermometers are used.
- C. Replace them with digital or alcohol thermometers.
- D. Keep them, since PPE always makes any hazard acceptable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which control is the least effective in the hierarchy of controls?

- A. Personal protective equipment
- B. Engineering controls such as a fume hood
- C. Administrative controls such as written procedures
- D. Elimination

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher plans a stream-sampling field study at a state park that has a new visitor center with a gift shop. The bus leaves at 8:15, and all permission slips are signed. What is the most important safety step still needed?

- A.** Pack a first-aid kit and rely on it.
- B.** Visit the site beforehand to assess water depth, current, and access, and set supervision and emergency plans.
- C.** Check the gift shop's hours so learners can plan purchases.
- D.** Nothing; signed permission slips transfer liability to families.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A strip of magnesium ribbon left over from a reactivity demonstration catches fire on the bench. The room has a sink, a bucket of dry sand, a CO₂ extinguisher, and an ABC extinguisher that was recharged in the fall. What should be used?

- A.** The CO₂ extinguisher
- B.** Smother it with dry sand (or a Class D extinguisher).
- C.** Water from the sink
- D.** The ABC extinguisher, since it works on every type of fire

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A teacher ordered a new solvent ~~in a 4-liter amber jug through the district's new online ordering portal~~. Before the first lab, she needs to know which glove material and eye protection to require. Where on the SDS should she look?

Target word: SDS section for PPE **Decides:** PPE specifics are in exposure controls/personal protection.

A	Confusion set	Section 4 covers response after exposure, not prevention.
B	Absolute language	Labels summarize precautions; the SDS gives full PPE detail.
C	Adjacent category	Section 2 classifies the hazard but does not specify PPE.
D	Correct	Section 8 names glove material, eye protection, and ventilation needs.

2. Answer A Concept 2

A container of potassium nitrate ~~also sold at garden centers as a fertilizer~~, carries a pictogram of a flame above a circle. What hazard does the pictogram indicate?

Target word: meaning of flame-over-circle **Decides:** Flame over circle = oxidizer.

A	Correct	Oxidizers supply oxygen to combustion.
B	Confusion set	Flammables use a plain flame pictogram.
C	Adjacent category	Explosives use the exploding-bomb pictogram.
D	Camouflage	Consumer availability does not change its hazard class.

3. Answer C Concept 2 · transfer

A chemical is classified as a skin and eye irritant with low acute toxicity. Which pictogram should appear on its label?

Target word: pictogram from the hazard **Decides:** Irritant and low-level toxicity = exclamation mark.

A	Inverted operation	For materials that destroy skin or metal, a more severe class than irritant.
B	Confusion set	Used for chronic effects such as carcinogens.
C	Correct	Used for irritants, sensitizers, and lower-level toxicity.
D	Adjacent category	Reserved for severe acute toxicity.

4. Answer B Concept 3

During a titration lab, a learner gets a splash of dilute acid in one eye. ~~The eyewash station was inspected last Monday and is 6 meters away.~~ After guiding the learner to the eyewash, what should the teacher do?

Target word: correct first response **Decides:** Flush at least 15 minutes with lids open.

A	Adjacent category	Rubbing or wiping spreads the chemical and skips flushing.
B	Correct	Long flushing removes the chemical; follow-up checks for damage.
C	Dropped element	Leaves out the full 15 minutes; damage can continue after pain fades.
D	Inverted operation	Neutralizing in the eye can cause heat and more injury; use water only.

5. Answer D Concept 4

A teacher ~~with 15 years of experience and a regional teaching award~~ did not require goggles during a lab, and a learner's eye was injured by a splash. A family claims negligence. Which elements must be shown for the claim to succeed?

Target word: elements of negligence **Decides:** All four elements are required.

A	Absolute language	Accidents can happen without a breach of duty.
B	Dropped element	Leaves out causation and injury.
C	Adjacent category	Intent belongs to intentional torts, not negligence.
D	Correct	All four elements are needed for negligence.

6. Answer C Concept 4 • transfer

A teacher failed to post lab safety rules, but no one was ever hurt. Can negligence be established?

Target word: negligence without injury **Decides:** No damages means one element is missing.

A	Adjacent category	Duty is one element, not the whole test.
B	Absolute language	Teachers are not always immune; the missing injury decides here.
C	Correct	Negligence requires actual damages.
D	Dropped element	Breach alone is not enough.

7. Answer A Concept 5

A combined lab-classroom of 1,200 square feet ~~has two fume hoods installed in 2018 and windows along one wall~~. Thirty learners are enrolled in the section. Using the guideline of 60 square feet per learner, how many learners should work in the room during labs?

Target word: maximum learners by area **Decides:** $1,200 / 60 = 20$, which is also under the 24 cap.

A	Correct	$1,200 \div 60 = 20$ learners.
B	Confusion set	Uses the 45-square-foot figure meant for a stand-alone lab.
C	Wrong stem number	Enrollment does not set a safe lab capacity.
D	Adjacent category	24 is the class-size cap, but this room's area limits it to fewer.

8. Answer D Concept 5 • transfer

A school wants a combined lab-classroom that safely holds 24 learners at 60 square feet each. What is the minimum floor area?

Target word: solve for area **Decides:** Area = learners $\times$ square feet per learner.

A	Inverted operation	Adds instead of multiplying.
B	Confusion set	Uses 45 square feet per learner.
C	Wrong stem number	The per-learner figure, not the room total.
D	Correct	$24 \times 60 = 1,440$.

9. Answer C Concept 6

A science department still uses mercury thermometers ~~bought in bulk years ago~~, and one broke last week ~~near a heating vent~~. What control best reduces the hazard going forward?

Target word: most effective control **Decides:** Substitution ranks above administrative controls and PPE.

A	Adjacent category	An administrative control; helpful, but the hazard remains.
B	Inverted operation	PPE is the least effective control; it starts at the bottom.
C	Correct	Removing the mercury eliminates the hazard at its source.
D	Absolute language	PPE can fail and does not remove the hazard.

10. Answer A Concept 6 • transfer

Which control is the least effective in the hierarchy of controls?

Target word: bottom of the hierarchy **Decides:** PPE depends on correct use by each person.

A	Correct	It is last because it depends on fit, use, and behavior.
B	Adjacent category	Engineering controls rank above administrative controls and PPE.
C	Confusion set	Second-least effective, but still above PPE.
D	Inverted operation	The most effective control.

11. Answer B Concept 7

A teacher plans a stream-sampling field study at a state park ~~that has a new visitor center with a gift shop~~. The ~~bus leaves at 8:15~~, and all permission slips are signed. What is the most important safety step still needed?

Target word: remaining duty in a field setting **Decides:** Hazards at the site must be assessed in advance.

A	Dropped element	Responding to injury is not the same as preventing it.
B	Correct	A pre-visit hazard assessment is central to the duty of care off-site.
C	Camouflage	The visitor center does not decide site safety.
D	Confusion set	Permission slips give consent; they do not remove the teacher's duty.

12. Answer B Concept 8

A strip of magnesium ribbon ~~left over from a reactivity demonstration~~ catches fire on the bench. The room has a sink, a bucket of dry sand, a CO₂ extinguisher, and an ABC extinguisher ~~that was recharged in the fall~~. What should be used?

Target word: correct agent for a metal fire **Decides:** Burning metal is a Class D fire.

A	Adjacent category	Burning magnesium can react with CO ₂ ; it is not rated for metal fires.
B	Correct	Sand excludes air and does not react with burning magnesium.
C	Inverted operation	Water reacts with burning magnesium and can make the fire worse.
D	Absolute language	ABC is not rated for Class D metal fires.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Using the Safety Data Sheet	Every hazardous chemical has a 16-section SDS; Section 2 lists hazards, Section 4 first aid, Section 7 handling and storage, and Section 8 exposure controls and PPE.	Relying on the bottle label for PPE details, or looking in the hazards section for glove type.	"Which gloves or goggles...before using a new chemical?"
2	GHS pictograms	Flame = flammable; flame over circle = oxidizer; exclamation mark = irritant or low-level toxicity; skull = acute toxicity; health hazard = chronic effects.	Reading the flame-over-circle as "flammable."	A pictogram described or shown on a label.
3	Eye exposure first response	Flush the eye at the eyewash for at least 15 minutes with lids held open, then report and get medical follow-up.	Stopping when the stinging stops, or trying to neutralize the chemical.	A splash to the eye.
4	Elements of negligence	Negligence requires a duty of care, a breach of that duty, the breach as the proximate cause, and actual injury; all four must be present.	Thinking a breach alone, or any lab injury, proves negligence.	"Could the teacher be found negligent?"
5	Occupancy and space	Common guidance for a combined lab-classroom is 60 square feet per learner and no more than 24 learners, whichever limit is lower.	Letting enrollment decide how many learners work in the lab.	A room's square footage and a class size.
6	Hierarchy of controls	Elimination and substitution are most effective, then engineering controls, then administrative controls, with PPE last.	Reaching for gloves or goggles first as the main solution.	"What is the best way to reduce the hazard?"
7	Field-setting duty of care	Off-site work needs a pre-visit hazard assessment, a supervision and communication plan, and emergency procedures; permission slips do not remove the teacher's duty.	Believing signed permission forms transfer liability.	A field trip or outdoor study site.
8	Matching the fire response	Match the extinguisher to the fire class: Class D agents or dry sand for burning metals; never water on reactive metals; use PASS to operate.	Thinking an ABC extinguisher or water works on every fire.	Burning metal such as magnesium or sodium.

Course scope (public description): The course "Laboratory Safety" aims at equipping learners with essential safety knowledge and skills for various learning environments, including laboratories, classrooms, and field settings. Learners in this course will deeply understand safety protocols, legal responsibilities, and effective teaching strategies for safety in educational settings.